

# BUDGET IMPACT ANALYSIS

## Ferric Derisomaltose for Heart Failure in a Rapidly Aging Regional Population



KEMENTERIAN KESIHATAN MALAYSIA  
HOSPITAL TAIPING

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### KEY

Introducing Ferric Derisomaltose therapy in Perak public hospitals is projected to generate **RM 1.3 million** in net savings over 3 years.

#### 1 INTRODUCTION

Heart failure with reduced ejection fraction (HFrEF) is a major burden in Malaysia, especially in the **rapidly aging state of Perak** (11.0% elderly vs 6.7% nationally). Iron deficiency affects 61.4% of this cohort and independently increases hospitalization risk.

Although guidelines recommend intravenous **ferric derisomaltose (FDI)**, its economic viability within Perak’s publicly funded MOH system was previously undefined.

**OBJECTIVE** Estimate the 3-year (2026–2028) budget impact of introducing FDI for HFrEF patients with ID in Perak from the MOH perspective.

#### 2 MATERIALS AND METHODS

- Model structure:** Excel-based BIA comparing a “World-with-FDI” with standard care (no IV iron).
- Cohort sizing:** Top-down funnel using Perak demographics, MY-HF registry, Asian-HF data and 74.7% public-sector utilization.
- Clinical inputs:** RR 0.83 for hospitalization; rolling-cohort dosing with high-dose loading and low-dose maintenance, derived from IRONMAN.
- Economic inputs:** RM 6,205 annual inpatient cost per person (MOH multi-centre data); public procurement drug pricing.

#### 3 RESULTS · BASE CASE

Phased uptake of 10% → 20% → 30% results in 1,761 patients receiving FDI by 2028.

**17%**  
REDUCTION IN  
HOSPITALIZATIONS

**RM 1.3M**  
3-YEAR CUMULATIVE  
SAVINGS

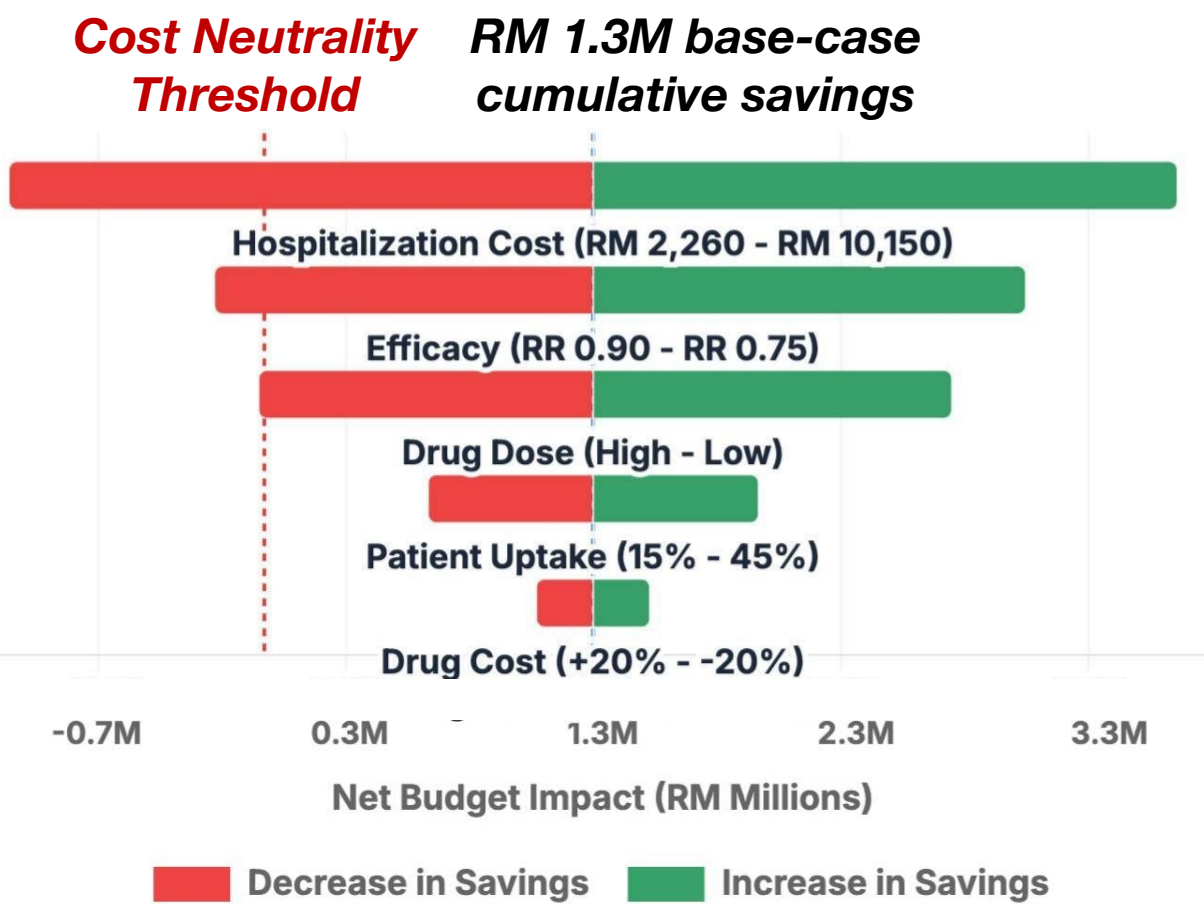
#### INVESTMENT-TO-RETURN TRAJECTORY

- Year 1:** RM 2,686 net investment. Front-loaded 1,978 mg loading doses are almost entirely offset by avoided admissions.
- Years 2–3:** Low-dose maintenance drives RM 442,279 savings in Y2 and RM 889,529 in Y3 as clinical benefits outpace acquisition costs.

| NET BUDGET IMPACT       | 2026          | 2027            | 2028            |
|-------------------------|---------------|-----------------|-----------------|
| Drug cost (RM)          | 618,719       | 792,951         | 968,061         |
| Hospitalization saving  | (616,033)     | (1,235,230)     | (1,857,590)     |
| <b>NET SAVINGS (RM)</b> | <b>–2,686</b> | <b>+442,279</b> | <b>+889,529</b> |

#### 4 RESULTS · SENSITIVITY ANALYSIS

One-way sensitivity analyses show the model remains **cost-saving across uptake and drug-cost variations**. The unit cost of hospitalization is the principal value driver.



*Tornado diagram above illustrating the sensitivity of the net budget impact to key model parameters.*

**COST NEUTRALITY THRESHOLD** RR 0.89.  
An 11.0% or greater real-world reduction in hospitalizations projects net budget savings for the state.

#### 5 DISCUSSION / CONCLUSION

FDI appears to be potentially **cost-saving intervention** for HFrEF with iron deficiency in Perak, challenging the view of IV iron as a pure cost driver.

#### STRATEGIC IMPLEMENTATION

- Patient selection:** Prioritizing high-utilization patients with recent admissions can amplify savings to >RM 3.6M. Low-cost stable cohorts may produce a deficit.
- Operational efficiency:** A single high-dose visit (up to 1,000 mg in 15-30 minutes) reduces pressure on crowded day-care centres versus multi-dose formulations.

**Integrating FDI gives the MOH a financially sustainable pathway to align with international standards of care.**

#### 6 LIMITATIONS

Relies on extrapolated demographic trends and national cost averages (RM6,205); excludes indirect societal costs and non-linear market uptake realities.

### REFERENCES

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Full References

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